



Frame 0



Frame 20



Frame 59

1

Physics Rollout

MPM forward (T steps)

Optimize $\delta \mathbf{F}_c$

Inject $\partial \mathcal{L} / \partial \mathbf{F}$

Output: $\mathbf{x}_T, \mathbf{F}_T$

2

Multi-view Render

$\mathbf{F} \rightarrow \mathbf{R}, \mathbf{S}$ (polar)

$\Sigma = \mathbf{S} \Sigma_0 \mathbf{S}^\top$

3DGS (8 views)

DT + IoU + Depth

$\rightarrow \partial \mathcal{L} / \partial \mathbf{F}$

3

Chamfer Plasticity

NN disp. $\mathbf{d}_i$ to target

KNN diffusion

$\delta \mathbf{F}_p$ from Jacobian

$\mathbf{F}_p \leftarrow (\mathbf{I} + \eta \delta \mathbf{F}_p) \mathbf{F}_p$

Rest $\rightarrow$ target

$\partial \mathcal{L} / \partial \mathbf{F}$ feedback (next episode)

● x: physics only

● Σ : shaped by $\mathbf{F}$

● $\mathbf{F}_p$: Chamfer plasticity